

Explainable Logistics Decision Intelligence

Predict delivery risk. Anticipate demand. Make scarce refrigeration an explicit business decision.

WHY ALL 85 ORDERS CANNOT BE DELIVERED

Refrigeration is a physical bottleneck



4 refrigerated vehicles × two full-volume trips. This optimistic bound ignores packing, time, weight and access limits.

9.229 m³

shortfall before the other constraints

79

orders served

6

deferred

0%

solver gap

Proof: the separate maximum-count solve also certifies 79. Every allocation passes the official checker. Optimality covers the implemented competition rules.

01 / PREDICT DELIVERY RISK

3.75 min MAE · 0.975 AUC

1,017 of 5,014 test orders flag at ≥50% late risk.
Action: review planned slack and handling buffers before dispatch.

Six-week test horizon; risks are predictions.

02 / ANTICIPATE DEMAND

Week 15 peak: 2,007 m³

Including 682 m³ chilled. **Action:** plan stock and refrigerated availability ahead of the ten-week forecast.



03 / ALLOCATE CAPACITY

79 served; 140.723 m³ chilled

Action: retain the balanced dispatch, explain each deferral, and choose explicitly whether backlog fairness should outweigh throughput.

Peak-day scenario; whole-order decisions.

THE COST OF CHANGING POLICY

Policy	Served	Chilled m³	Points*	Deferred	Backlog**
Fresh-first	79	140.723	35,770	6	9/10
Fairness-first	77	126.254	34,900	8	10/10
Balanced	79	140.723	35,770	6	9/10

*Common balanced weights. **Previously deferred orders served. Hard rules unchanged.

HONEST EVALUATION

Total-demand WAPE: **6.08%**; chilled: **5.24%**. Tech total-demand WAPE: **24.7% Kandy, 29.8% Peliyagoda**. Several per-series R² values are negative.

Nine simple Tech baselines on earlier chronological folds give modest/inconsistent gains. Verified predictions retained; final holdout excluded from that experiment.